

GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER VI

Name of Department: - Computer Science and Engineering

1. Subject Code: **TCS 514** Course Title: **Computer Networks - I**
2. Contact Hours: L: **3** T: **0** P: **0**
3. Examination Duration (Hrs): Theory **3** Practical **0**
4. Relative Weight: CIE **25** MSE **25** ESE **50**
5. Credits: **3**
6. Semester: **VI**
7. Category of Course: **DSC**
8. Pre-requisite: Fundamental of Computer & Introduction to Programming (TCS 101), Data Structures with C (TCS 302)

9. Course Outcome:	After completion of the course the students will be able to: CO1 Familiarize with fundamental underlying principles of computer networking and functionality of layered network architecture. CO2 Evaluate and Select the most suitable Application Layer protocol (such as HTTP, FTP, SMTP, DNS, BitTorrent) as per the requirements of the network application and work with available tools to demonstrate the working of these protocols CO3 Model a problem or situation in terms of layering concept and map it to the TCI/IP stack CO 4 Design a Reliable Data Transfer Protocol and incrementally develop solutions for the requirements of Transport Layer CO5 Articulate, analyze and appraise the TCP protocol CO6 Describe the essential principles of Network Layers, design the network layer requirements using both IPv4 and IPv6 and understand the supporting IP protocols
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10. Details of the Course:

Sl. No.	Contents	Contact Hours
1	Unit 1: Introduction: Computer Networks and the Internet, Overall view: As components and as services; What is a	8

	protocol, What is a network protocol, Access Networks and Physical Media, Circuit and Packet Switching, Internet Backbone, ISPs and Interconnection of ISPs, Delay, Loss, and Throughput in Packet-Switched Networks The Layered Architecture: Protocol Layering, The OSI Reference Model and the TCP/IP protocol stack	
2	Unit 2: Application Layer1: Principles and Architectures of Network Applications, Client and Server processes, the idea of socket, Transport services available to Application layer Application Layer Protocols: The Web and http: Persistent and Nonpersistent connections, http message format, User-Server interaction: cookies, proxy server, conditional GET Email: smtp, mail message formats, accessing mails using http and imap DNS: Services, How it works, Root, Top-Level and Authoritative DNS servers, DNS caching, Recursive and Iterative queries, DNS Records and messages P2P architecture: Scalability, BitTorrent protocol	9
3	Unit 3: Application Layer2: Video Streaming and Content Distribution Networks, HTTP Streaming and DASH. Transport Layer1: Introduction, Relationship between Transport Layer and Network Layer, Multiplexing and Demultiplexing work of Transport Layer, Connectionless Transport: UDP- Segment Structure of UDP, Checksum	8
4	Unit 4: Transport Layer2: Principles of Reliable Data Transfer, Building a reliable data transfer protocol using FSM, stop-and-wait, Go Back N, Selective Repeat Connection Oriented Transport TCP: Connection Establishment, TCP segment structure in detail, Round Trip Time Estimation and Timeout, TCP Flow Control, TCP Congestion Control: Slow start, Congestion Avoidance, Fast Recovery Introduction to QUIC: Quick UDP Internet Connections	9
5	Unit 5: Network Layer I: Introduction, Network Service Model, The Internet Protocol(IP), IPv4 Datagram format, DHCP, Network Address Translation (NAT), A brief introduction to routing protocols, RIP and OSPF. Network Programming: Sockets-Address structures, TCP sockets, creating sockets, bind, listen, accept, fork and exec function, close function; TCP client server: Echo server,	11

	Elementary UDP sockets: UDP echo server	
	Total	45

Textbooks:

Authors Name	Title	Edition	Publisher, Country	Year
Jim Kurose and Keith Ross	Computer Networking- A Top Down Approach	8th Edition	Pearson Education	2022

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Andrew Tanenbaum, Nick Feamster and David J. Wetherhall	Computer Networks	6 th Edition	Pearson Education	2022
Behrouz A. Forouzan and Firouz Mosharraf	Computer Networks: A Top-Down Approach	Standard Edition	McGraw Hill	2023

